EXPERIMENT 12

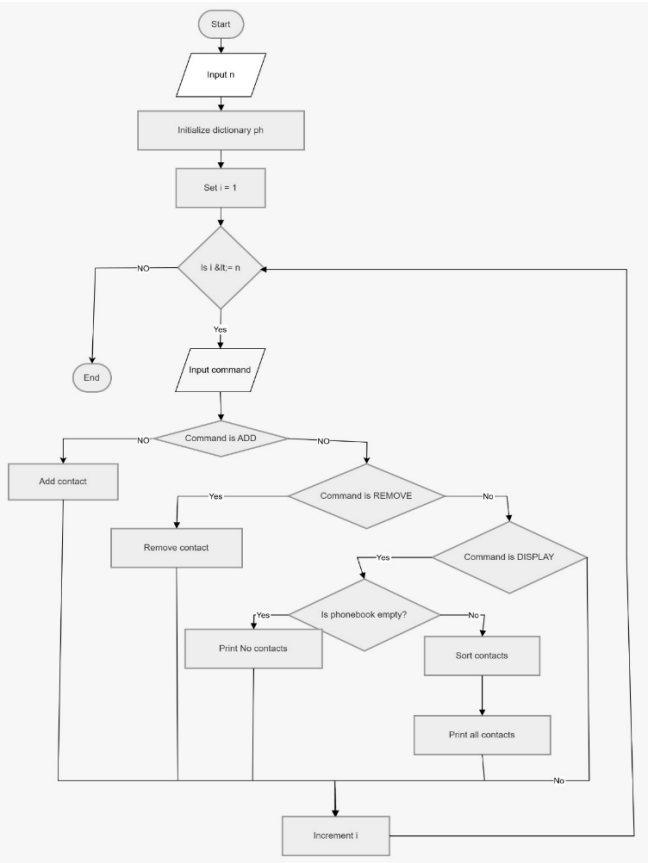
12.1.1//Phone num e database management

Ans: ALGORITHM:

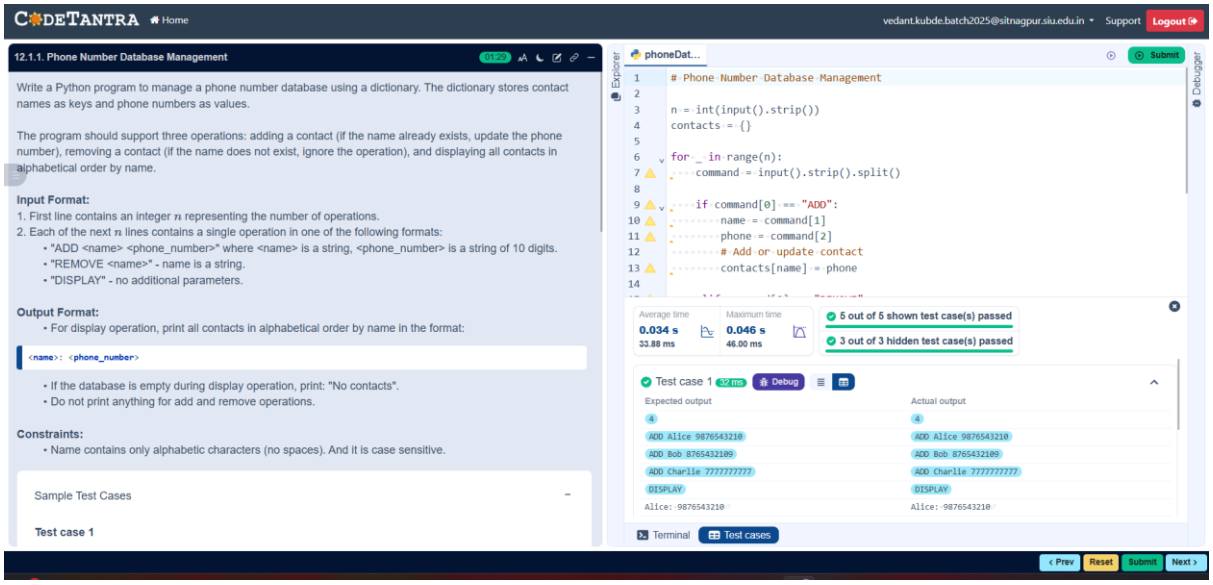
Step 1: Start
Step 2: Read integer n (number of commands)
Step 3: Create an empty dictionary ph
Step 4: Repeat steps for i = 1 to n

- Read command input and split into list cmd
- If cmd[0] == "ADD"
    - Add contact → ph[name] = number
- Else if cmd[0] == "REMOVE"
    - Remove contact using pop()
- Else if cmd[0] == "DISPLAY"
    - If dictionary is empty → Print **"No contacts"**
    - Else → Sort dictionary keys and print **name:number**
      Step 5: Stop

FLOWCHART:


Start
Input n
Initialize dictionary ph
Set i = 1
Is i <= n
NO
Yes
End
Input command
Command is ADD
NO
NO
Add contact
Command is REMOVE
Yes
No
Remove contact
Command is DISPLAY
Yes
No
Is phonebook empty?
Yes
No
Print No contacts
Sort contacts
Print all contacts
No
Increment i

CODETANTRA:


CODETANTRA
Home
vedant.kubde.batch2025@sitnagpur.siu.edu.in
Support
Logout
12.1.1. Phone Number Database Management
01:29
Write a Python program to manage a phone number database using a dictionary. The dictionary stores contact names as keys and phone numbers as values.
The program should support three operations: adding a contact (if the name already exists, update the phone number), removing a contact (if the name does not exist, ignore the operation), and displaying all contacts in alphabetical order by name.
Input Format:
1. First line contains an integer n representing the number of operations.
2. Each of the next n lines contains a single operation in one of the following formats:
• "ADD <name> <phone_number>" where <name> is a string, <phone_number> is a string of 10 digits.
• "REMOVE <name>" - name is a string.
• "DISPLAY" - no additional parameters.
Output Format:
• For display operation, print all contacts in alphabetical order by name in the format:
<name>: <phone_number>
• If the database is empty during display operation, print: "No contacts".
• Do not print anything for add and remove operations.
Constraints:
• Name contains only alphabetic characters (no spaces). And it is case sensitive.
Sample Test Cases
Test case 1
Explorer
phoneDat...
Submit
Debugger
# Phone Number Database Management
n = int(input().strip())
contacts = {}
for _ in range(n):
command = input().strip().split()
if command[0] == "ADD":
name = command[1]
phone = command[2]
# Add or update contact
contacts[name] = phone
Average time
0.034 s
33.88 ms
Maximum time
0.046 s
46.00 ms
5 out of 5 shown test case(s) passed
3 out of 3 hidden test case(s) passed
Test case 1
32 ms
Debug
Expected output
Actual output
4
ADD Alice 9876543210
ADD Bob 8765432109
ADD Charlie 7777777777
DISPLAY
Alice: 9876543210
Terminal
Test cases
Prev
Reset
Submit
Next